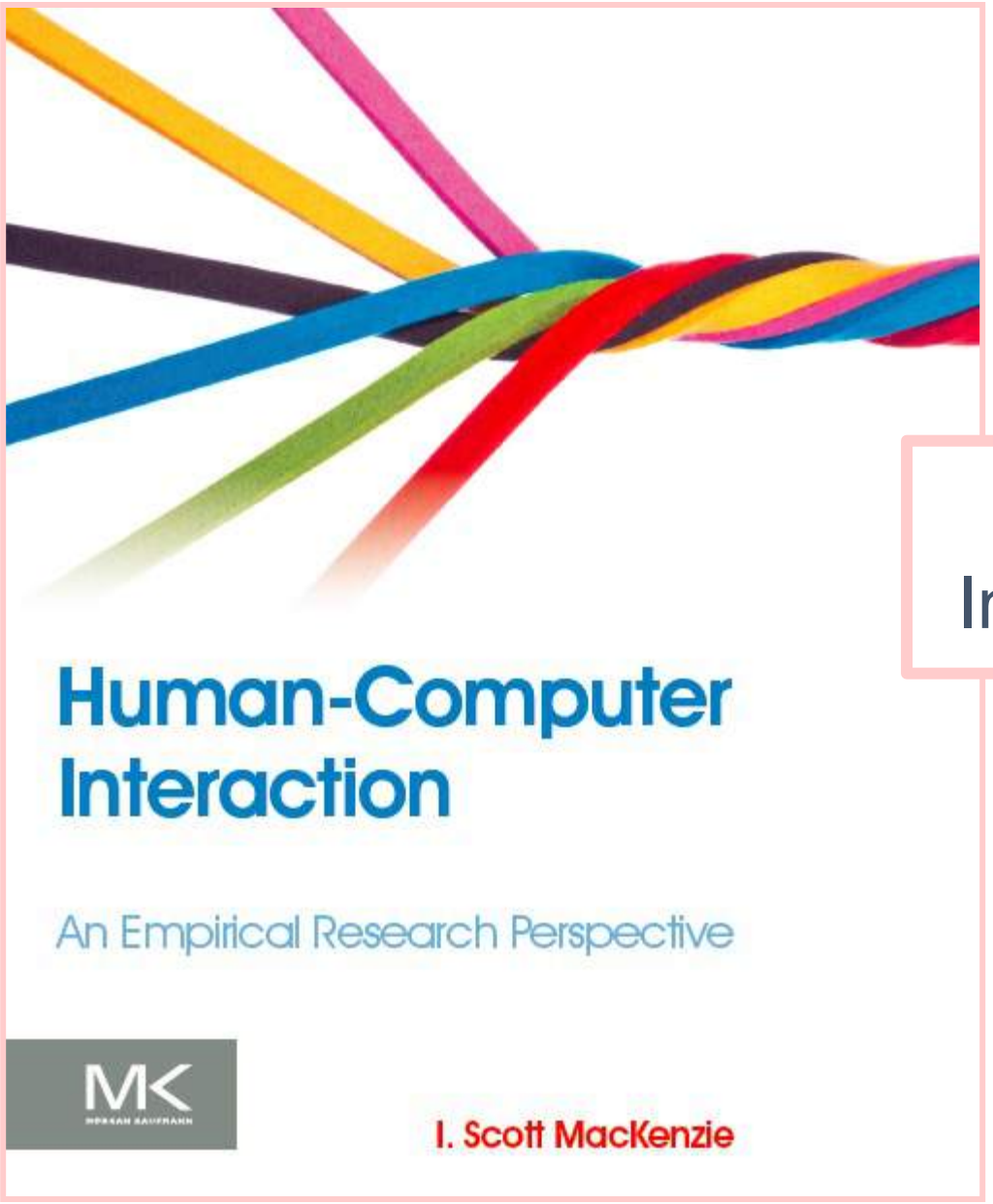



# Chapter 3
# Interaction Elements

**Longbiao CHEN**
**2018/04 Xiamen University**

# Interaction

- *Interaction* occurs when a human performs a task using computing technology

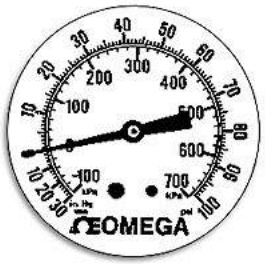
- Interaction tasks with a goal:
  - Send an e-mail
  - Burn a CD
  - Program a thermostat
  - Enter a destination in a GPS device

# Interaction Elements

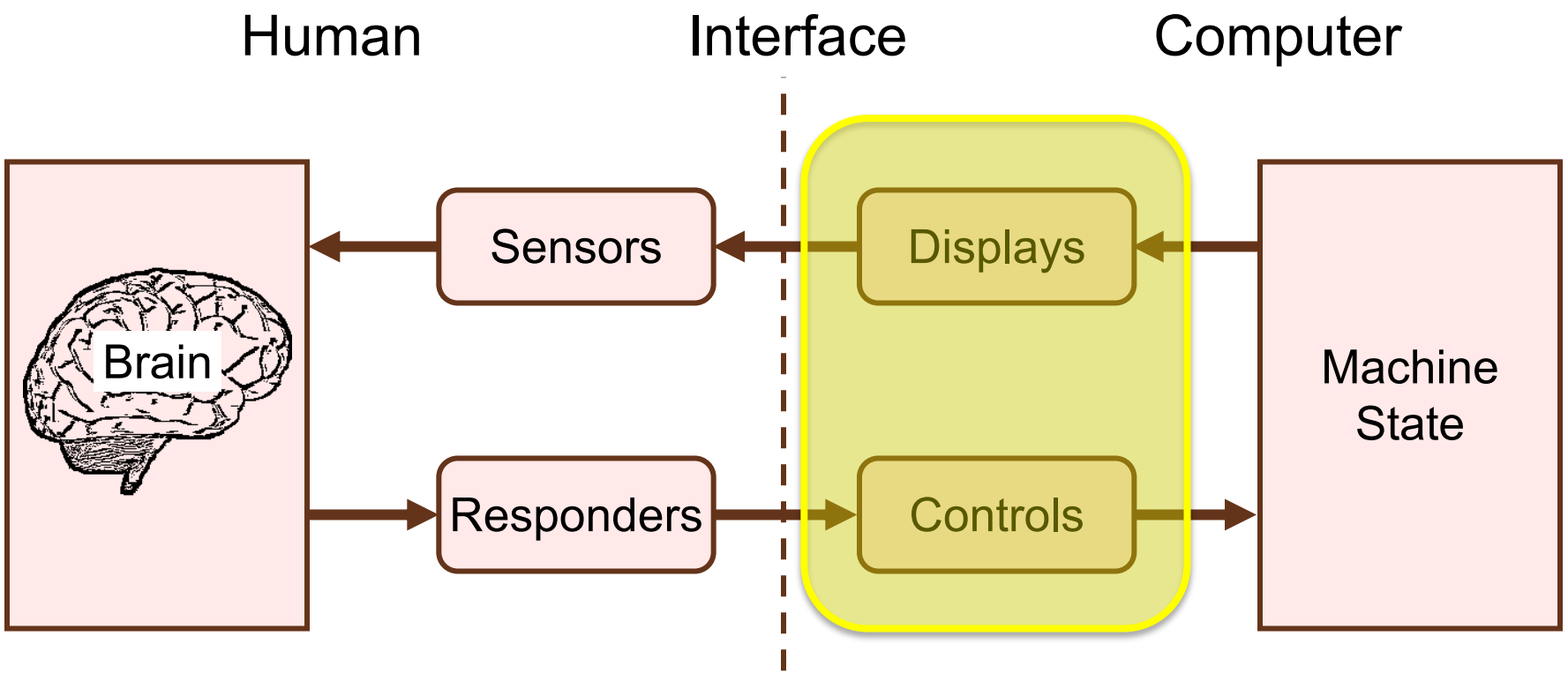
- Studied at many levels and in different contexts

- Early human factors research on “knobs and dials” is relevant today [旋钮和转盘]
  - Knobs → “controls”; dials → “displays”



# Human Factors Model (revisited)



# Hard Controls, Soft Controls

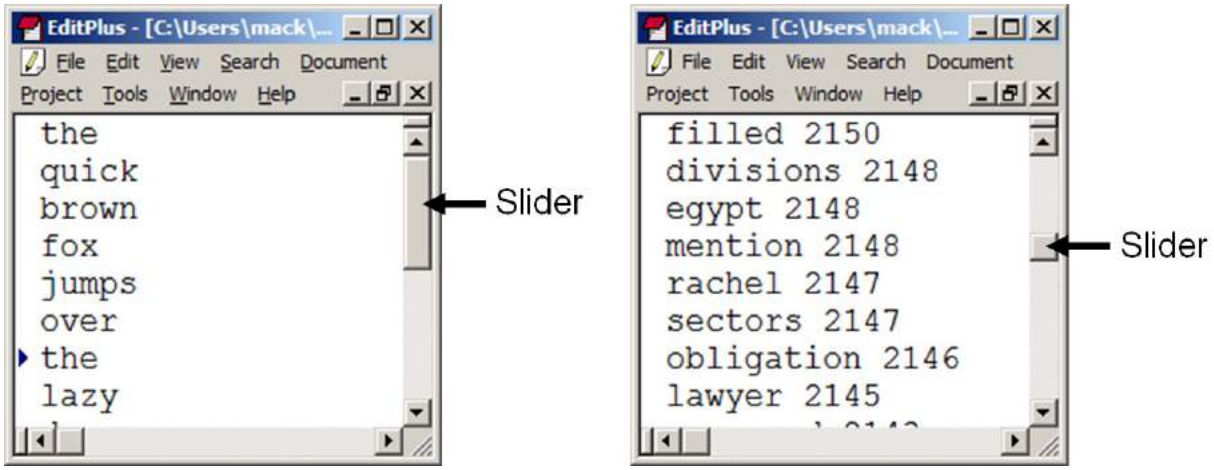
- In the past, controls were physical, single-purpose devices → *hard controls*
- Today's graphical displays are malleable [可重塑]
- Interfaces created in software → *soft controls*
- Soft controls rendered on a *display*
  - Distinction blurred between soft controls and displays




Soft controls are also displays!

# Scrollbar Slider

- Example of a soft control (control + display)
- As a control…
  - Moved to change view in document
- As a display…
  - Size reveals view size relative to entire document
  - Position reveals view location in document

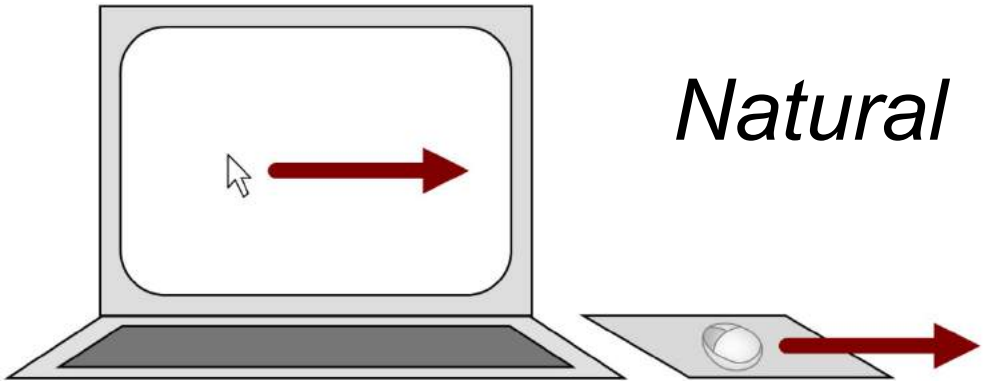


# Control-Display Relationships

- Also called *mappings*
- Relationship between operation of a control and the effect created on a display
- At least three types:
  - Spatial relationships
  - Dynamic relationships
  - Physical relationships

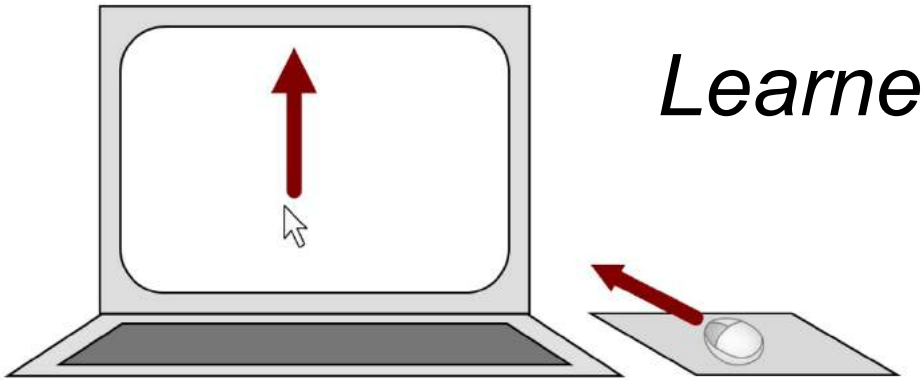
**[Fitts's Law]**

# Spatial Relationships



*Natural*

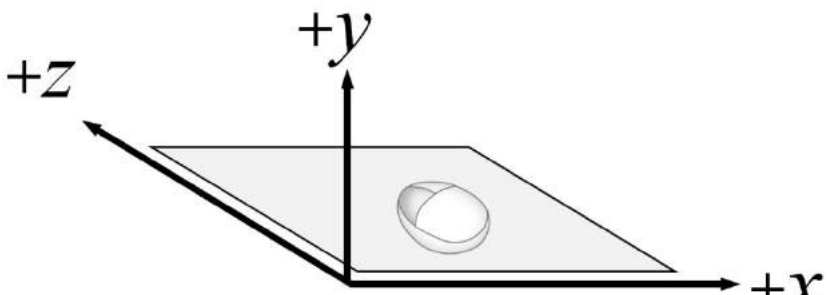
**Spatial congruence**
Control: right
Display: right



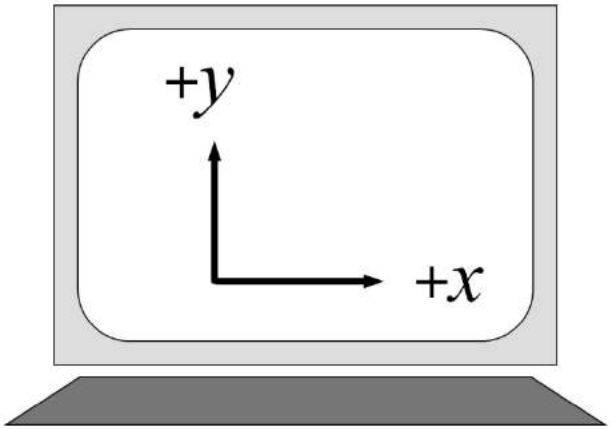
*Learned*

**Spatial transformation**
Control: forward
Display: up

# Axis Labeling



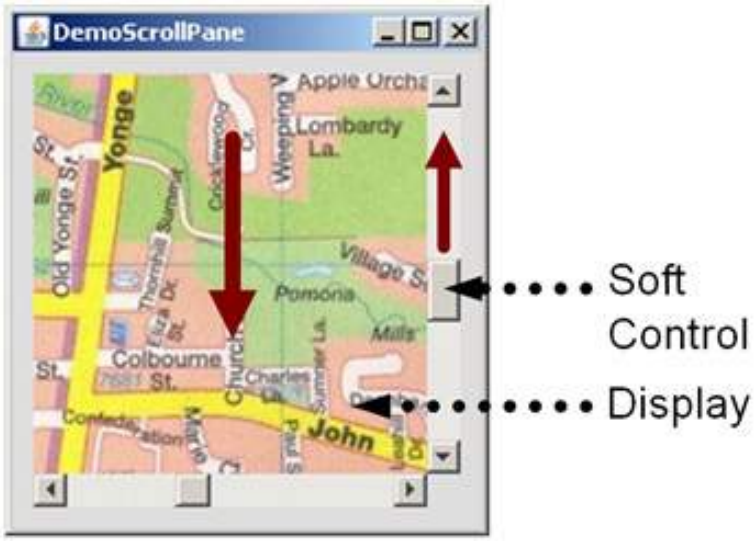

Control



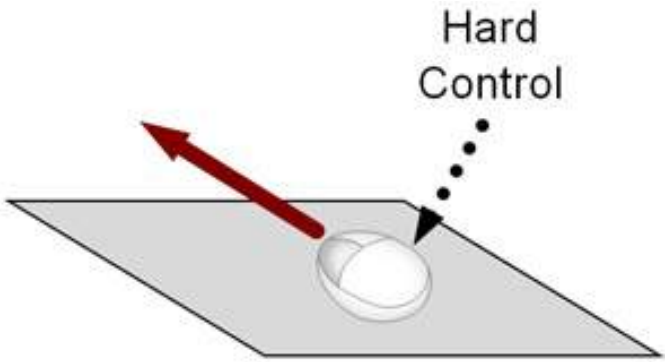

Display

| Axis | Control (mouse) | Display (cursor) |
|---|---|---|
| x | + | + |
| y | | + |
| z | + | |

# Third Tier



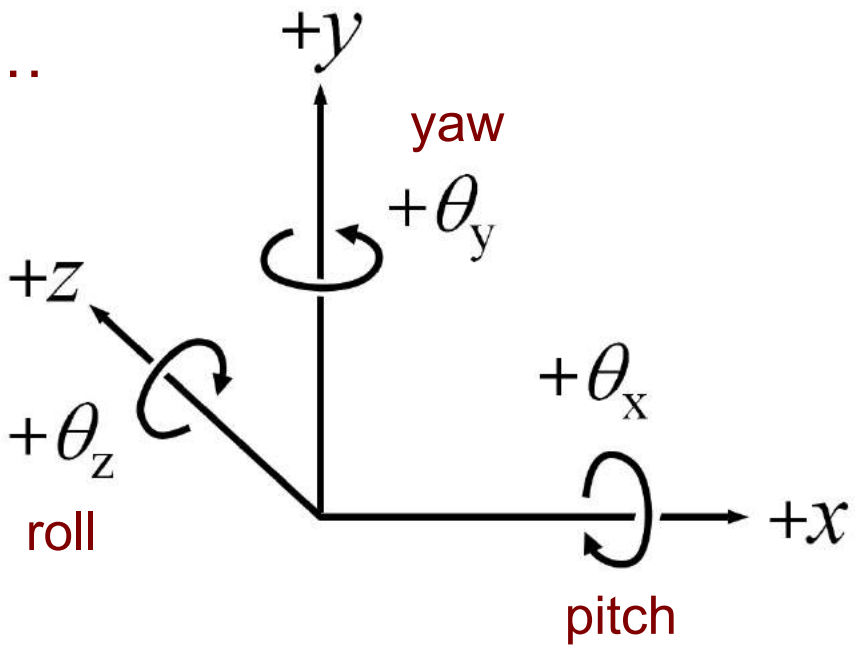

| DOF | Hard Control | Soft Control | Display |
|---|---|---|---|
| x | | | |
| y | | + | - |
| z | + | | |
| $\theta_x$ | | | |
| $\theta_y$ | | | |
| $\theta_z$ | | | |



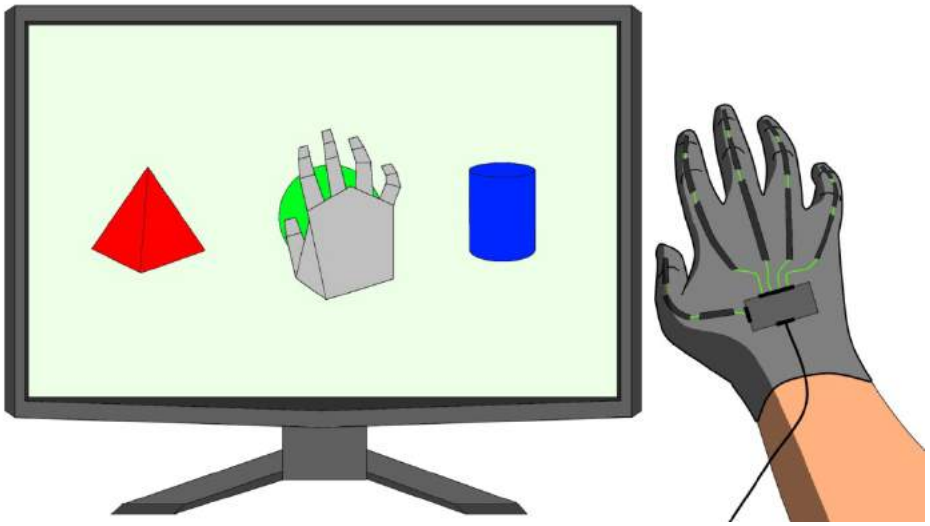
# Degree of Freedom

- In 3D there are 6 degrees of freedom (DOF) [自由度]
  - 3 DOF for position ($x$, $y$, $z$)
  - 3 DOF for orientation ($\theta_x$, $\theta_y$, $\theta_z$)

In aeronautics…



# Spatial Congruence in 3D

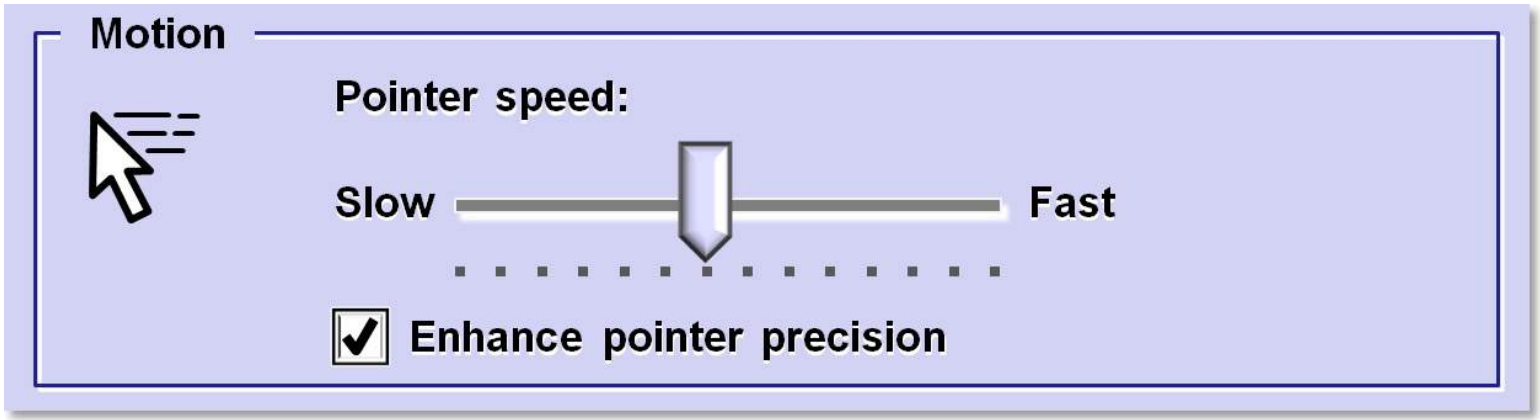


| DOF | Control | Display |
|---|---|---|
| x | + ● | ● + |
| y | + ● | ● + |
| z | + ● | ● + |
| $\theta_x$ | + ● | ● + |
| $\theta_y$ | + ● | ● + |
| $\theta_z$ | + ● | ● + |

[DynaShape: Mouse Interaction]

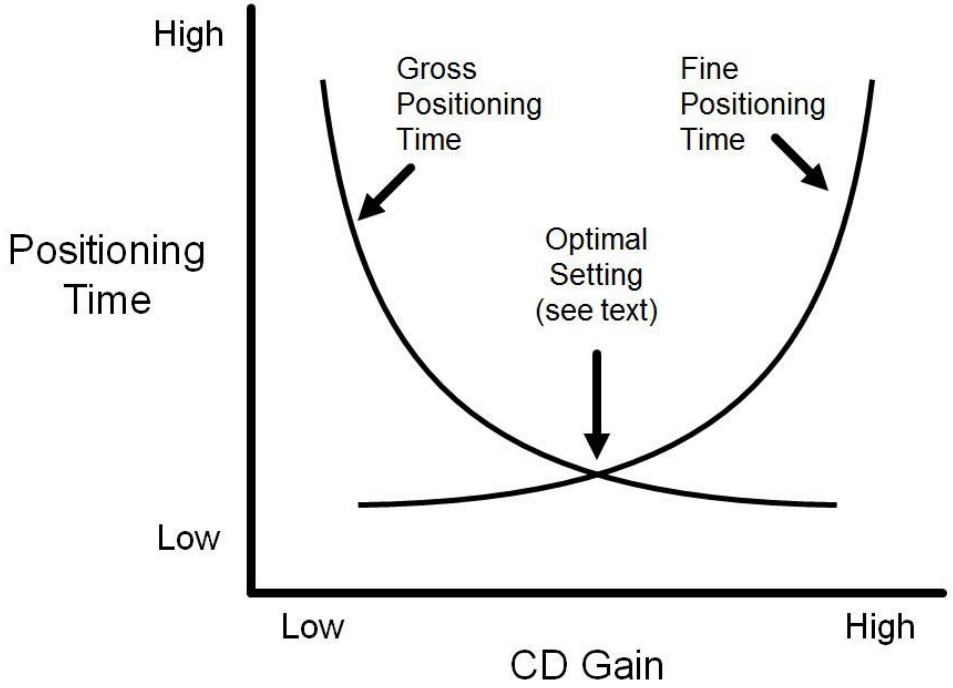
# Control-Display Gain (CD Gain)

- Quantifies the amount of *display* movement for a given amount of *controller* movement
- E.g., CD gain = "2" implies 2 cm of controller movement yields 4 cm of display movement
- Sometimes specified as a ratio (C:D ratio)
- For non-linear gains, the term *transfer function* is used
- Typical control panel to adjust CD gain:



# CD Gain and User Performance

- Tricky to adjust CD gain to optimize user performance
- Issues:
  - Speed accuracy trade-off (what reduces positioning time tends to increase errors)
  - Opposing relationship between gross and fine positioning times:



# Latency [延迟]

- *Latency* (aka *lag*) is the delay between an input action and the corresponding response on a display
- Usually negligible on interactive systems
  - e.g., cursor positioning, editing
- May be "noticeable" in some settings; e.g.,
  - Remote manipulation
  - Internet access (and other "system" response situations)
  - Virtual reality (VR)
- Human performance issues for empirical research

# VR Controllers

- 6 DOF controllers common in VR and other 3D environments
- Considerable processing requirements
- Lag often an issue
- E.g., [Polhemus G4]™ (see below)
- Lag specified as <10 ms (which is low)
- But the user experiences the complete system



# Property Sensed, Order of Control

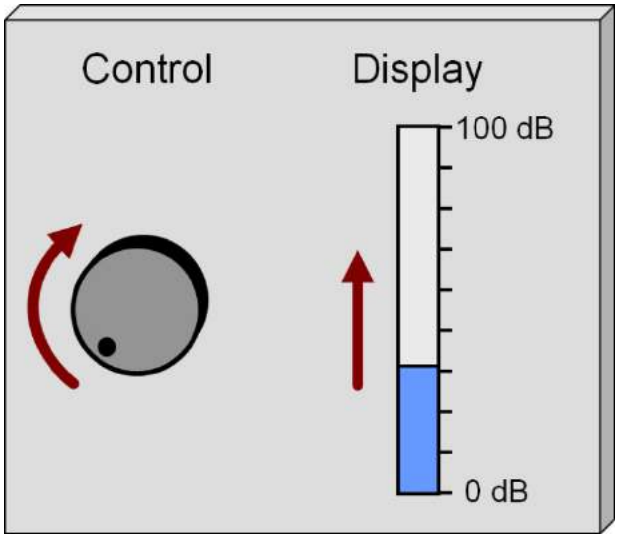
- Property sensed
  - Position (graphics tablet, touchpad, touchscreen)
  - Displacement (mouse, joystick)
  - Force (joystick)
- Order of control
  - Position (of cursor/object)
  - Velocity (of cursor/object)

# Natural vs. Learned Relationships

- Natural relationships → spatially congruent
- Learned relationships → spatial transformation

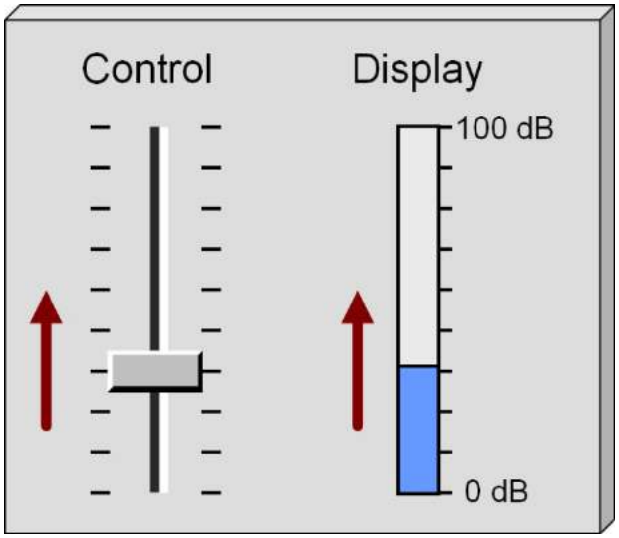
Examples

Learned relationship




| DOF | Control | Display |
|---|---|---|
| x | | |
| y | | + |
| z | | |
| $\theta_x$ | | |
| $\theta_y$ | | |
| $\theta_z$ | + | |

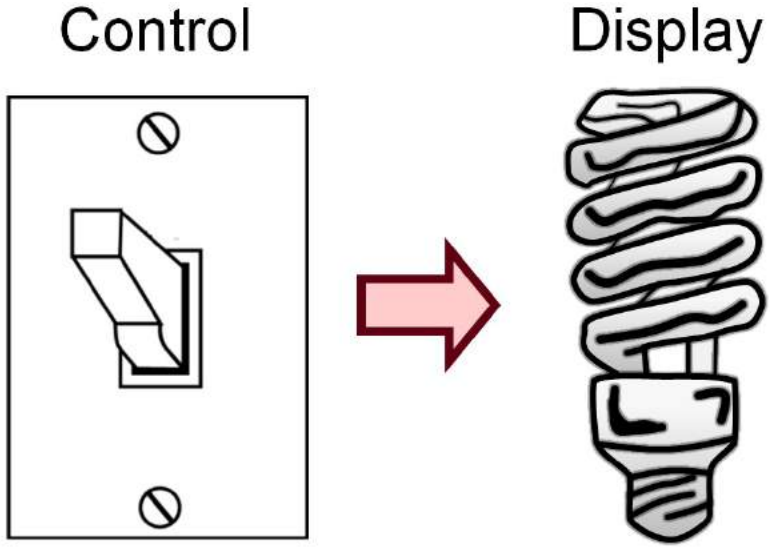
Natural relationship




| DOF | Control | Display |
|---|---|---|
| x | | |
| y | + | + |
| z | | |
| $\theta_x$ | | |
| $\theta_y$ | | |
| $\theta_z$ | | |

# Learned Relationships

- Learned relationships seem natural if they lead to a *population stereotype* or *cultural standard*
- A control-display relationship needn't be a spatial relationship…

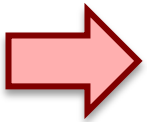



Is the display on or off?

Answer: On (in U.S., Canada)
Off (in U.K.)

# Mental Models

- Related: *physical analogy*, *metaphor*, *conceptual models*
- Definition: a physical understanding of an interface or interaction technique based on real-world experience
- Scroll pane: slider up, view up ("up-up" is a conceptual model that helps our understanding)
- [Desktop metaphor] is most common metaphor in computing
- Other commonly exploited real-world experiences:
  - Shopping, driving a car, calendars, painting
- [Icon design], in general, strives to foster mental models



# Graphics and Paint Applications

- Icons attempt to leverage real-world experiences with painting, drawing, sketching, etc.



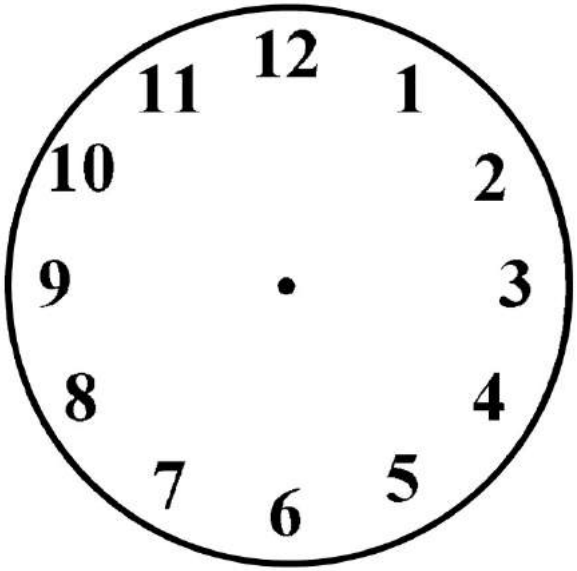




tooltips help for obscure features

# Clock Metaphor

- Numeric entry on PDA[1]
- Users make straight-line strokes in direction of digit on clock face





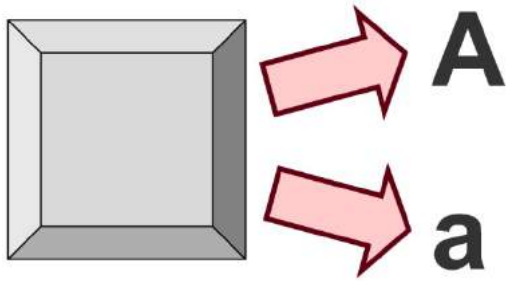





[1] McQueen, C., MacKenzie, I. S., & Zhang, S. X. (1995). An extended study of numeric entry on pen-based computers. *Proceedings of Graphics Interface '95*, 215-222, Toronto: Canadian Information Processing Society.

# Modes

- A *mode* is a functioning arrangement or condition
- Modes are everywhere (and in most cases are unavoidable)
- Office phone light: *on* = message waiting, *off* = no messages
- Computer keyboards have modes
  - ≈100 keys + SHIFT, CTRL, ALT ➔ ≈800 key variations



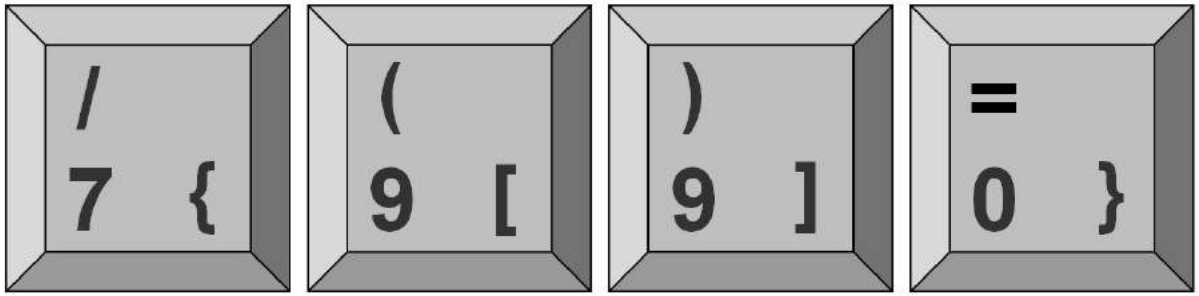
# F9 – Microsoft Word (2010)

- At least six interpretations, depending on mode:

| | |
|---|---|
| F9 → | Update selected fields |
| SHIFT+F9 → | Switch between a field code and its result |
| CTRL+F9 → | Insert an empty field |
| CTRL+SHIFT+F9 → | Unlink a field |
| ALT+F9 → | Switch between all field codes and their results |
| ALT+SHIFT+F9 → | Run GOTOBUTTON or MACROBUTTON from the field that displays the field results |

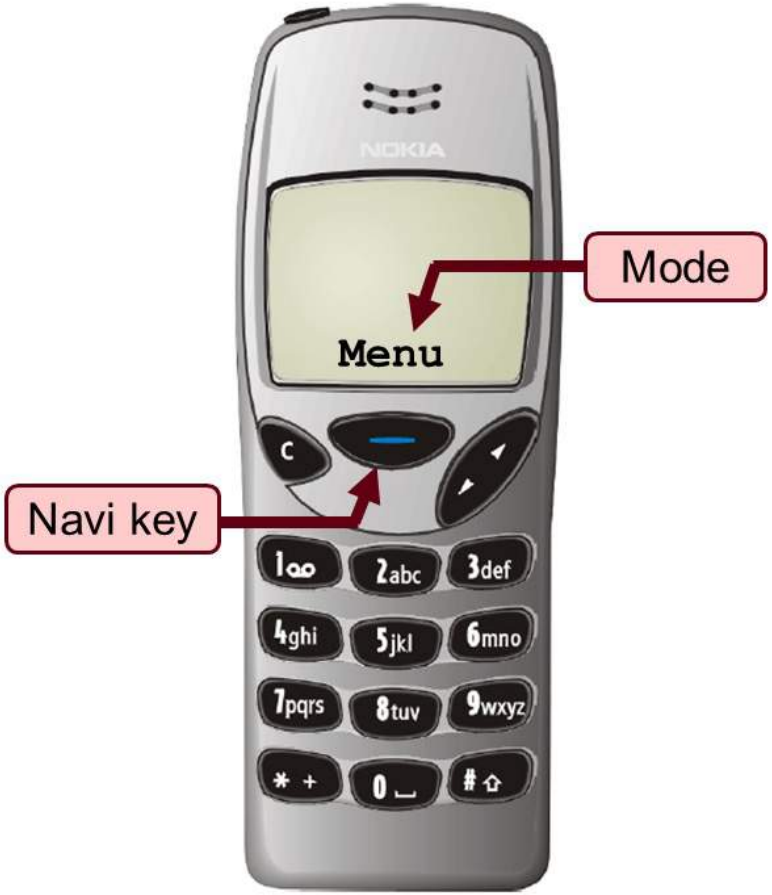
# International Keyboards

- Some keys bear three symbols
- How to access the third symbol?
- German keyboard example:



# Mobile Phone Example

- Navi key (first introduced on Nokia *3210*)
- Mode revealed by word above
- At least 15 interpretations: Menu, Select, Answer, Call, End, OK, Options, Assign, Send, Read, Use, View, List, Snooze, Yes




[Nokia 3210]

# Contemporary LCD Monitor

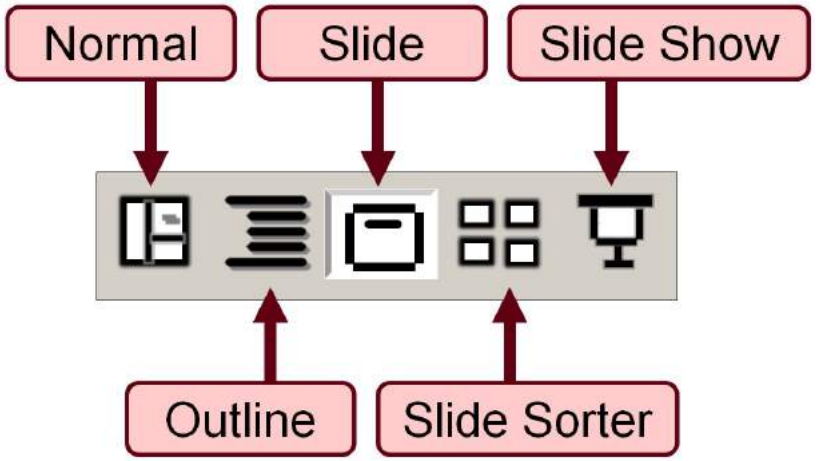
- Similar to Navi key idea
- No labels for the four buttons above power button
- Function revealed on display when button pressed
- Possibilities explode




[Arduino LCD Menu]

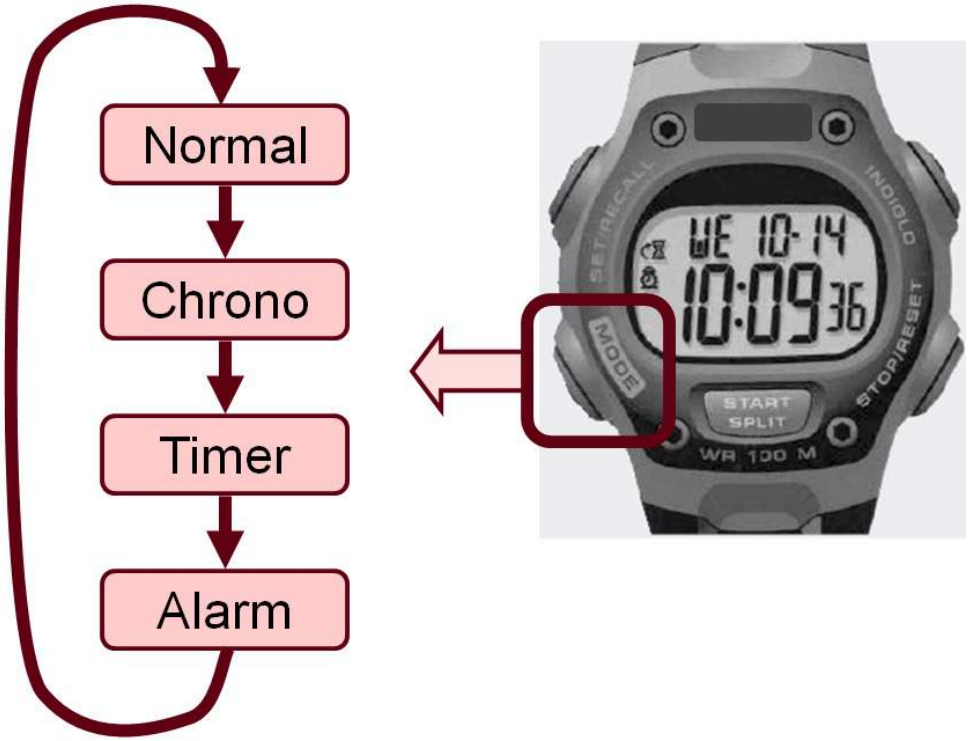
# Mode Switching

- PowerPoint: Five view modes
- Switch modes by clicking soft button
- Current mode apparent by background shading
- Still, problems lurk
- How to exit Slide Show mode?
  - PowerPoint → ESC
  - Firefox → ?



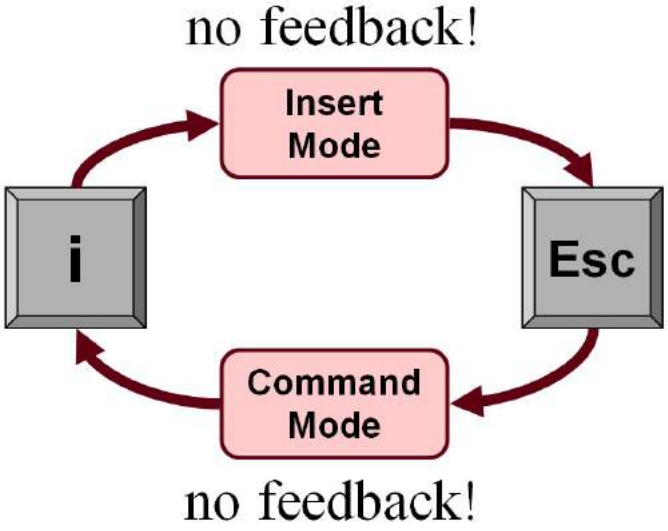
# Mode Switching (2)

- Sports watch
- Single button cycles through modes



# Mode Visibility

- Shneiderman: “offer information feedback”[1]
- Norman: “make things visible”[2]
- unix *vi* editor: Classic example of no mode visibility:

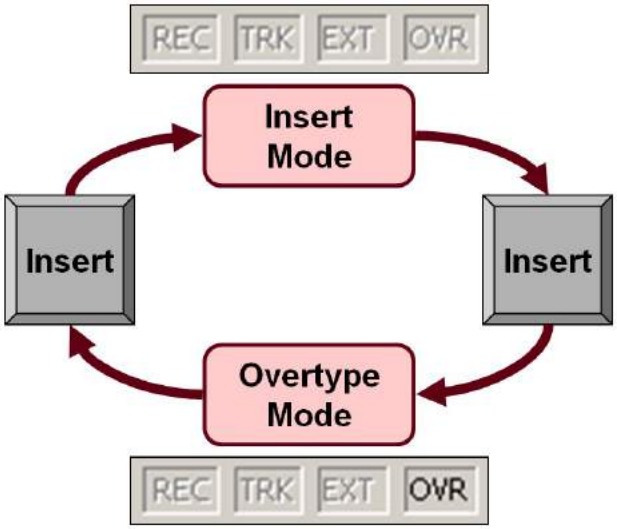



[1] Shneiderman, B., & Plaisant, C. (2005). *Designing the user interface: Strategies for effective human-computer interaction.* (4th ed.). New York: Pearson.

[2] Norman, D. A. (1988). *The design of everyday things*. New York: Basic Books.

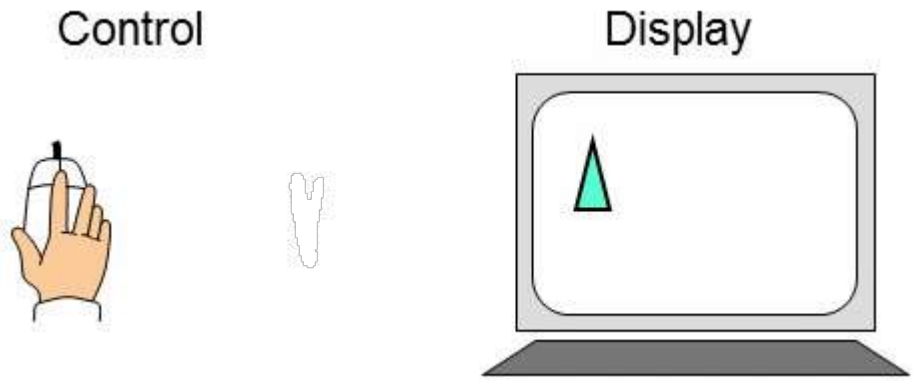
# Mode Visibility (2)

- Insert vs. Overtype mode on MS/Word
- Some variation by version, but the user is in trouble most of the time



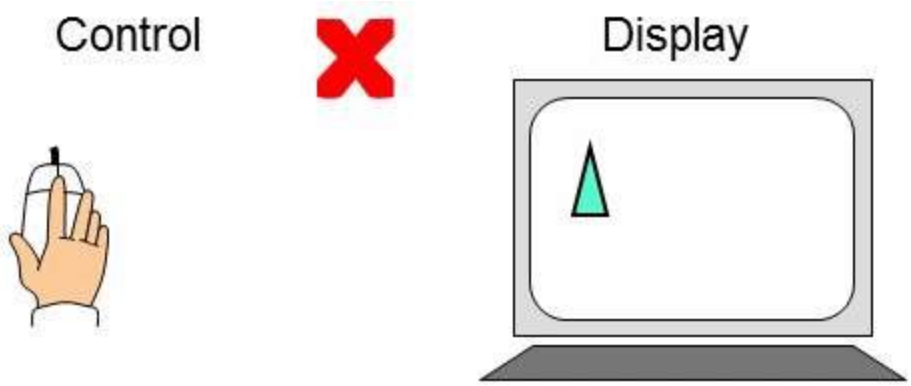
# Modes and Degrees of Freedom

- If control DOF < display DOF, modes are necessary to fully access the display DOF
- Consider a mouse (2 DOF) and a desktop display (3 DOF)
- *x*-*y* control (no problem):



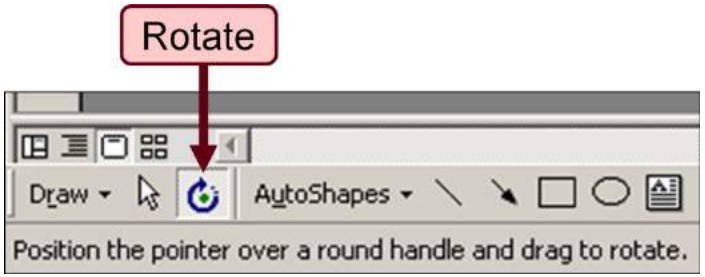

but…

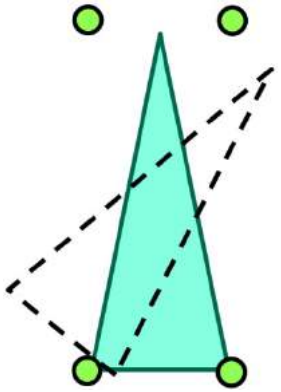
- Rotation is a problem:



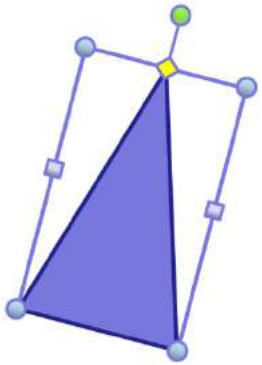
# Rotate Mode

- The solution: Rotate mode
- Two approaches
  - Separate rotate mode:



  - Embedded rotate mode:



Could be avoided with…

# 3 DOF Mouse

- Lots of research:

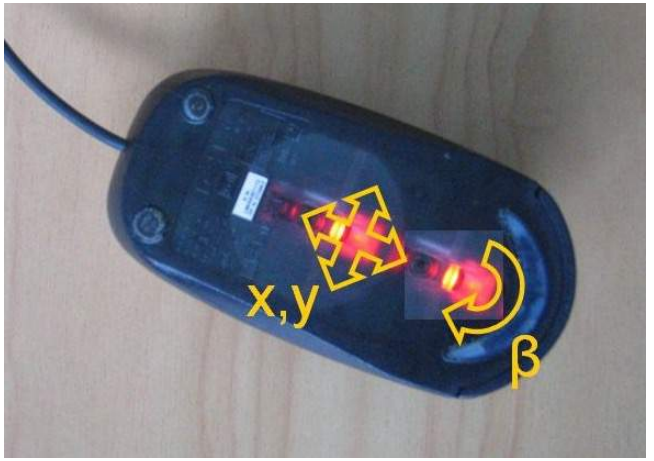
1



2



3




- But no commercial products (yet!)

[1] MacKenzie, I. S., Soukoreff, R. W., & Pal, C. (1997). A two-ball mouse affords three degrees of freedom. *Proc CHI '97*, 303-304, New York: ACM.

[2] Almeida, R., & Cubaud, P. (2006). Supporting 3D window manipulation with a yawing mouse. *Proc NordiCHI 2006*, 477-480, New York: ACM.

[3] Hannagan, J., & Regenbrecht, H. (2008). *TwistMouse for simultaneous translation and rotation*. Tech Report. HCI Group. Information Science Department. University of Otago, Dunedin, New Zeland.

# A Two-Ball Mouse Affords Three Degrees of Freedom

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**ABSTRACT**

We describe a prototype two-ball mouse containing the electronics and mechanics of two mice in a single chassis. Unlike a conventional mouse, which senses *x*-axis and *y*-axis displacement only, our mouse also senses *z*-axis angular motion. This is accomplished through simple calculations on the two sets of *x*-*y* displacement data. Our mouse looks and feels like a standard mouse, however certain primitive operations are performed with much greater ease. The rotate tool – common in most drawing programs – becomes redundant as objects are easily moved with three degrees of freedom. Mechanisms to engage the added degree of freedom and different interaction techniques are discussed.

**Keywords**

pointing devices, multi-degree-of-freedom input, rotation

**INTRODUCTION**

Since the introduction of the Apple *Macintosh* in 1984, the form of desktop systems has not changed substantially. Manipulating complex graphic objects usually combines mouse movement with special tools or modes. These permit simple displacement data from the mouse to map to and control the displacement and/or angular location of objects or scenes. These mappings are often unnatural.

**DIMENSIONS AND DEGREES OF FREEDOM**

Before we describe our mouse, it is important to distinguish between dimensions (D) and degrees of freedom (df). In three dimensions, there are six degrees of freedom: three for position along the *x*, *y*, and *z* axes, and three for angular orientation ($\theta x$, $\theta y$, and $\theta z$). In 2D, there are three degrees of freedom. If we consider a 2D surface such as a mouse pad (see Figure 1), then there is an *x* and *y* positional degree of freedom and a *z*-axis rotational degree of freedom.

Figure 2 illustrates the dimensions and degrees of freedom for several "mouse-type" devices. As a standard mouse is manoeuvred on a mouse pad, only its *x* and *y* displacement are sensed, as indicated in the "Mouse" column. This is sufficient for most tasks using, for example, word processors or spreadsheets. However, within drawing packages and other graphics programs, a common task is to move an object to a new location and with a new orientation. This is a full two dimensional task and it requires three degrees of freedom. Since mouse angular motion is not sensed, a rotate tool or a manipulator handle is usually required. Although other schemes can be devised to control the orientation of objects [3], they are unnatural and violate the basic perceptual structure of interaction [2].

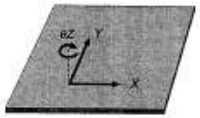



Figure 1. A mouse pad is a two-dimensional surface with three degrees of freedom: *x*, *y*, and $\theta z$.

| | Degree of Freedom | Mouse | 2-ball Mouse | Other A | B | C | D |
|---|---|---|---|---|---|---|---|
| 2D | *x* | ● | ● | ● | ● | ● | ● |
| | *y* | ● | ● | ● | ● | ● | ● |
| | θz | ○ | ● | ○ | ● | ○ | ● |
| 3D | *z* | ○ | ○ | ● | ● | ○ | ○ |
| | θx | ○ | ○ | ○ | ● | ● | ● |
| | θy | ○ | ○ | ○ | ● | ● | ● |

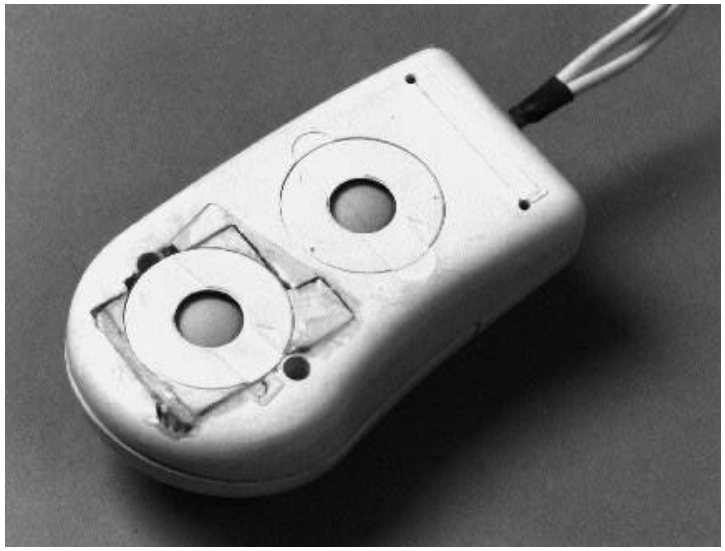
Figure 2. Dimensions vs. degrees of freedom. Grey dots indicate degrees of freedom sensed for several types of input devices. (See text for discussion)

**A TWO-BALL MOUSE**

We have built a prototype of a 2D/3df mouse (see Figure 3) that makes a rotate tool redundant. The mouse was built using the mechanical parts from two Microsoft 2.0 mice and the electronics from two Fellowes MousePens. The prototype weighs 156 g, compared to 104 g for the standard Microsoft 2.0 mouse. It interfaces to a host computer through two serial ports. This is sufficient for a prototype and it allowed a variety of simple demonstrations to be implemented. As a product, a complete bottom-up redesign would be required.

With some simple arithmetic in the interface software, the *z*-axis or rotational component of the mouse's motion is easily computed from the two streams of *x*-*y* positional





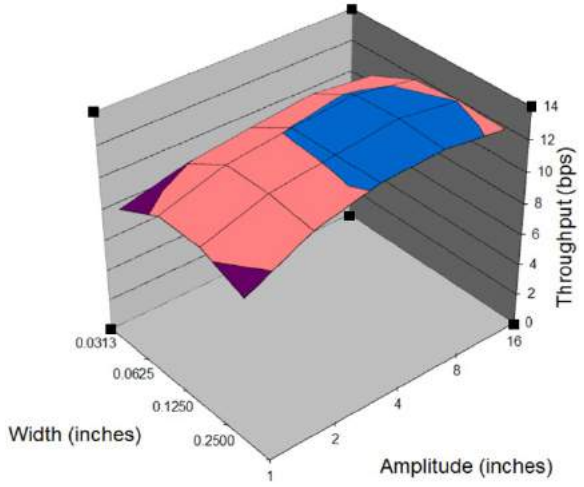


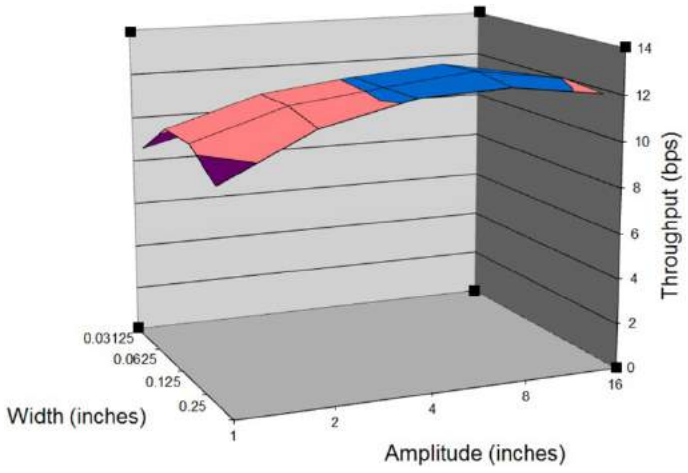
**YouTube video**

http://www.youtube.com/watch?v=nQvowU_gzpc&feature=youtu.be

# 3D Rotation

- Mapping controller *x*-*y* to display $\theta_x$-$\theta_y$-$\theta_z$
- Very awkward (to be polite)



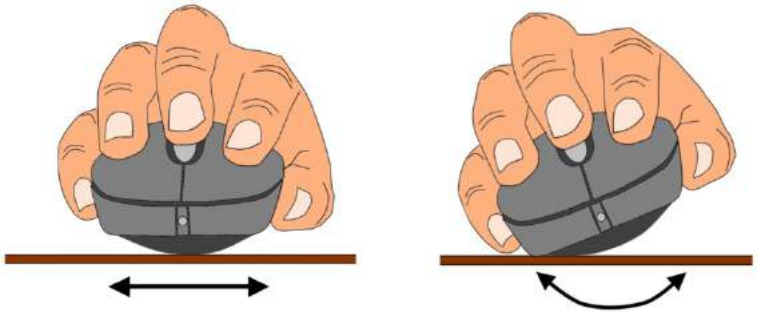

**Demo**

[click here] to open Excel spreadsheet

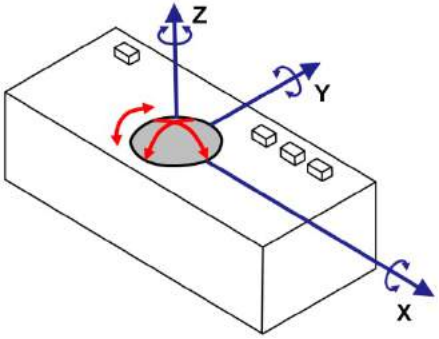
(3DRotationExample.xlsx)

# >2 Degrees of Freedom

- Examples in the HCI research literature
- 4 DOF *Rockin' Mouse*[1]



- Three-axis trackball[2]




[1] Balakrishnan, R., Baudel, T., Kurtenbach, G., & Fitzmaurice, G. (1997). The Rockin'Mouse: Integral 3D manipulation on a plane. *Proc CHI '97*, 311-318, New York: ACM.

[2] Evans, K. B., Tanner, P. P., & Wein, M. (1981). Tablet based valuators that provide one, two, or three degrees of freedom. *Computer Graphics, 15*(3), 91-97.

# Separating the Degrees of Freedom

- More DOF is not necessarily better
- Must consider the context of use
- Etch-A-Sketch: separate 1 DOF *x* and *y* controllers:



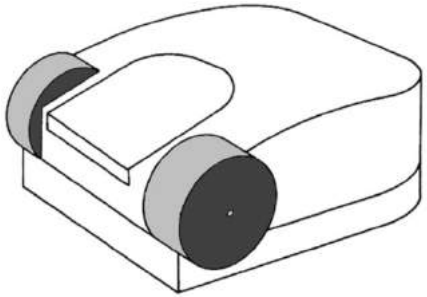
# Wheel Mouse

- Separate DOF via a wheel
- Successful introduction by Microsoft in 1996 with the *IntelliMouse* →

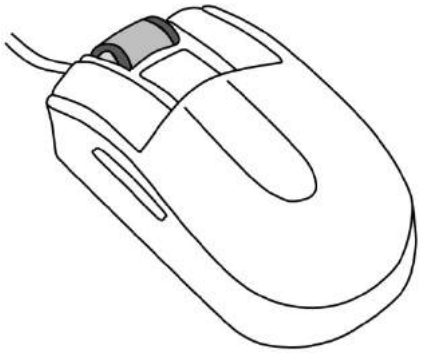



- Preceded by…

RollerMouse[1]



ProAgio[2]



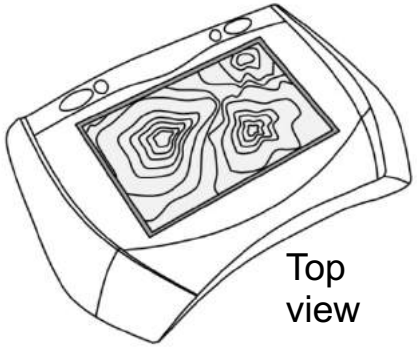
[1] Venolia, D. (1993). Facile 3D manipulation. *Proc CHI '93*, 31-36, New York: ACM.
[2] Gillick, W. G., & Lam, C. C. (1996). U. S. Patent No. 5,530,455.

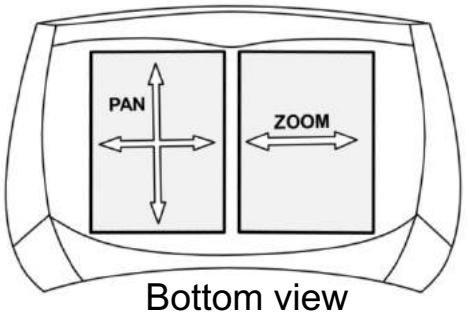
# Adding a Touch Sensor

PadMouse[1]



Panning and Zooming Display[2]




Multitouch+Mouse[3]



[1] Balakrishnan, R., & Patel, P. (1998). The PadMouse: Facilitating selection and spatial positioning for the non-dominant hand. *Proc CHI '98* (pp. 9-16): New York: ACM.

[2] Silfverberg, M., Korhonen, P., & MacKenzie, I. S. (2003). International Patent No. WO 03/021568 A1.

[3] Villar et al. (2009). Mouse 2.0: Multi-touch meets the mouse. *Proc UIST '09*, 33-42, New York: ACM.

# Mobile Context

- Touchscreens are the full embodiment of direct manipulation
- No need for a cursor (cf. indirect input)






# Touch Input Challenges

- Occlusion and accuracy ("fat finger problem")
- Early research → Offset cursor[1]
- Contemporary systems use variations; e.g., offset animation:







[1] Potter, R., Berman, M., & Shneiderman, B. (1988). An experimental evaluation of three touch screen strategies within a hypertext database. *Int J Human-Computer Interaction, 1* (1), 41-52.

# Multitouch



# Accelerometers

- Accelerometers enable tilt or motion as an input primitive
- Technology has matured; now common in mobile devices
- Many applications; e.g., spatially aware displays:



# Interaction Errors

- Discussions above focused on physical properties of controllers and the interactions they enable
- Interaction involves the human (sensors, brain, responders) and the machine
- Interaction errors are unavoidable (and, hence, are akin to an "interaction element")
- We conclude with a look at interaction errors and their consequences
- Themes: (see **HCI:ERP** for discussion)
  - Big, bad errors are high in consequences and therefore get a lot of attention
  - Little errors are low in consequences and therefore tend to linger
  - There is a continuum

# Discard Changes

- Default dialogs to quit an application:









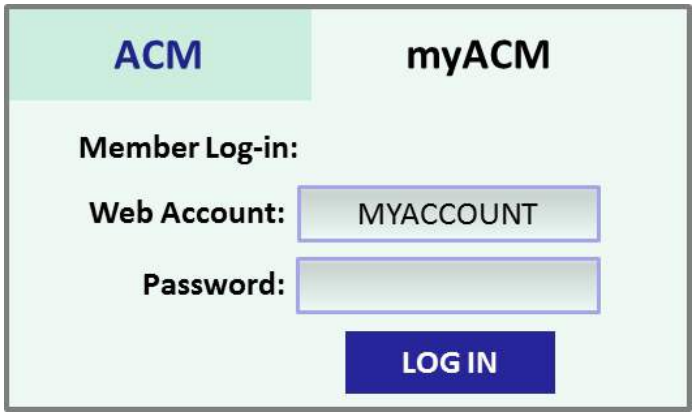


# CAPS_LOCK

- Some log-in dialogs alert the user if CAPS_LOCK is on…

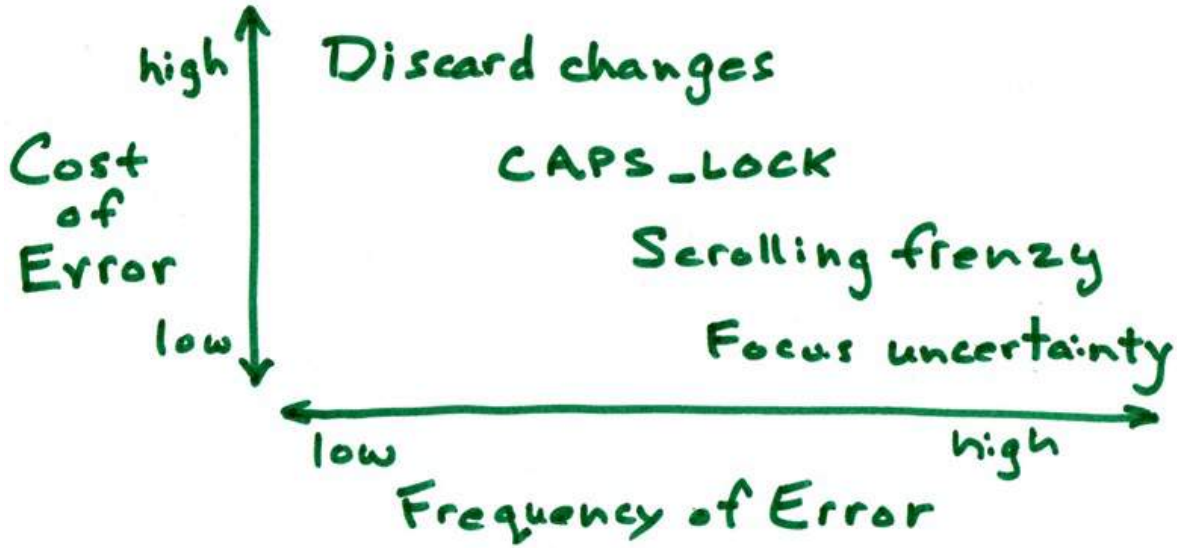



- while others do not…



# Cost vs. Frequency of Errors

- Message: High frequency / low cost errors are the most interesting
- They…
  - Have evaded the scrutiny of designers
  - Keep users on guard

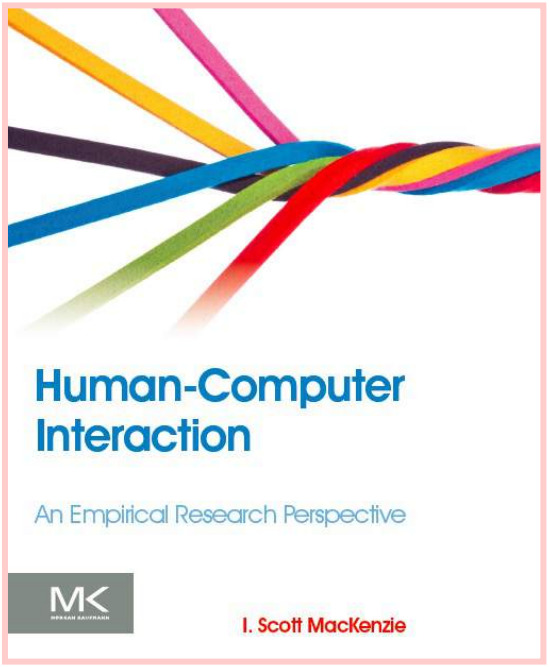


# How to Start A Boeing 737

[Cockpit Scenes - 737 Start Up]



# Thank You


Human-Computer Interaction
An Empirical Research Perspective
I. Scott MacKenzie